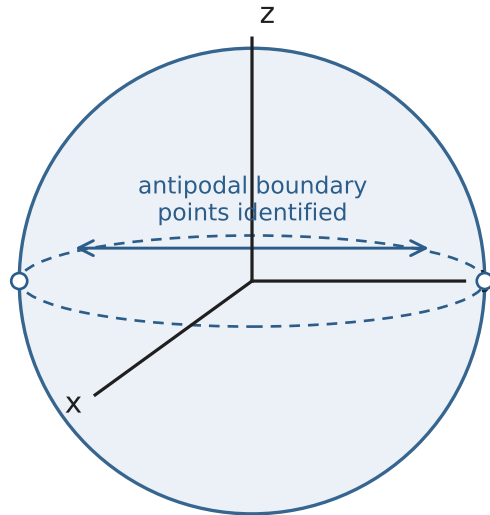


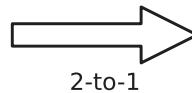
SU(2): The Double Cover of Spatial Rotations

SO(3) physical rotations



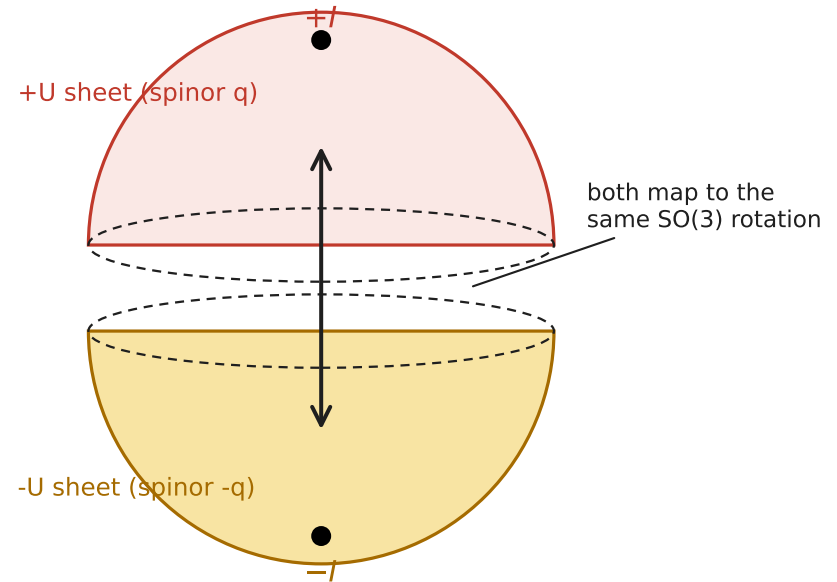
rotation ball: $|\omega| \leq \pi$

$$\pi: SU(2) \rightarrow SO(3)$$

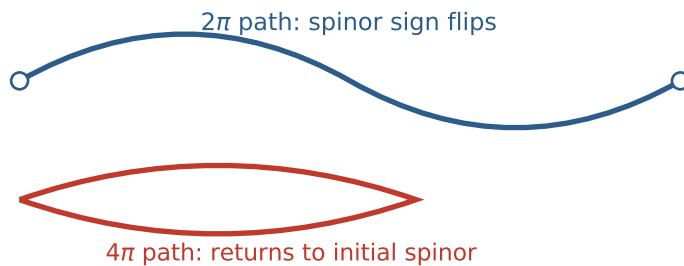


2-to-1

SU(2) spinor manifold



Representative paths



Topological readout

SU(2) contains two spinor states, q and $-q$, for each physical SO(3) rotation. A single full turn is not homotopic to the identity; two full turns are.

Fermions are SU(2) representations: spin-1/2 follows from the double cover.